🏆 Deep Learning Challenge

Smart Health Risk Predictor ML vs DL Comparative Study

📊 Dataset You Can Use :

https://www.kaggle.com/datasets/redwankarimsony/heart-disease-data

It is one of the most famous medical ML datasets, It predicts presence of heart diseas

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📌 Dataset Overview

Target column:

- `0` → No heart disease
- `1` → Heart disease present

🧾 Features Description

Feature	Description
age	Age of patient
sex	1 = male, 0 = female
cp	Chest pain type (4 categories)
trestbps	Resting blood pressure
chol	Serum cholesterol
fbs	Fasting blood sugar > 120 mg/dl
restecg	Resting ECG results
thalach	Maximum heart rate achieved
exang	Exercise induced angina
oldpeak	ST depression induced by exercise

Feature	Description
slope	Slope of ST segment
ca	Number of major vessels (0–3)
thal	Thalassemia (normal/fixed/reversible defect)

🎯 Challenge Objective

You must:

1. Build at least one **classical Machine Learning model**
2. Build one **Deep Learning model** using:
   - TensorFlow

     OR
   - PyTorch
3. Perform a **comparative study**
4. Analyze and interpret results

🧠 Required Steps

🔹 Step 1 : Data Understanding

- Explore dataset
- Check missing values
- Check class distribution
- Statistical summary

🔹 Step 2 : Data Preprocessing

They must:

- Handle missing values

- Encode categorical features
- Feature scaling (StandardScaler)
- Train/Test split
- Handle imbalance (optional bonus: SMOTE)

⚠️ Important:

DL models REQUIRE normalized input

🔹 Step 3 : Machine Learning Model

Minimum requirement:

Choose one:

- Logistic Regression
- Random Forest
- SVM
- XGBoost

Evaluate using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

🔹 Step 4 : Deep Learning Model

Build a Neural Network:

Example architecture:

Input Layer

→ Dense (64 neurons, ReLU)

→ Dense (32 neurons, ReLU)

→ Output Layer (Sigmoid)

Train for 30–100 epochs

Use:

- Binary Crossentropy loss
- Adam optimizer

Evaluate using same metric

🔹 Step 5 : Comparative Study

You must compare:

Criteria	ML Model	DL Model
Accuracy	?	?
Precision	?	?
Recall	?	?
F1-score	?	?
Training time	?	?
Overfitting behavior	?	?

And answer:

- Which model performed better?
- Why?
- Is DL always better than ML?
- When should we prefer ML over DL?

📦 Deliverables

Each team must submit :

1️⃣ Notebook (.ipynb)

Containing:

- Clean code
- Clear comments
- Visualizations
- Metrics

2️⃣ Readme file Report (Max 3 pages)

Must include:

- Introduction
- Methodology
- ML model results
- DL model results
- Comparative analysis
- Final conclusion

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